

Capstone Section 3 - MongoDB

express-app/ is the actual starter backend provided in the assignment (MongoDB.zip: server.js, models/, routes/, public/) - not custom-built. It connects to a local MongoDB 7.0.14 instance, database schoolSystem, and was run for real to produce the output below.

Setup

```
cd express-app
npm install
node server.js
```

Console output:

```
School API running at http://localhost:3001
Connected to MongoDB: schoolSystem
```

Seeded schools, courses and enrollments in MongoDB Compass / via mongosh with the sample data from the assignment brief (Greenwood High School, Riverside Public School, plus their courses and two student enrollments).

Task: create a new entry in school

Inserted via mongosh (equivalent to MongoDB Compass's "ADD DATA"), with the required fields `_id`, `name`, `address`, `principal`:

```
db = db.getSiblingDB("schoolSystem");
db.schools.insertOne({
  _id: ObjectId("665f1fa4a7d3f1a0aabc1003"),
  name: "Lakeside Academy",
  address: "789 Birch Lane, Lakeview",
  principal: "Dr. Maria Gomez"
});
```

Result:

```
Insert result:
{
  acknowledged: true,
  insertedId: ObjectId('665f1fa4a7d3f1a0aabc1003')
}
```

```
db.schools.find():
{
  _id: ObjectId('665f1fa4a7d3f1a0aabc1001'),
  name: 'Greenwood High School',
  address: '123 Maple Street, Springfield',
  principal: 'Mr. John Adams'
}
{
  _id: ObjectId('665f1fa4a7d3f1a0aabc1002'),
  name: 'Riverside Public School',
  address: '456 Oak Avenue, Riverdale',
  principal: 'Ms. Linda Carter'
```

```

}
{
  _id: ObjectId('665f1fa4a7d3f1a0aabc1003'),
  name: 'Lakeside Academy',
  address: '789 Birch Lane, Lakeview',
  principal: 'Dr. Maria Gomez'
}

```

RESTful API (from the provided express-app)

Verified against the running server:

```
$ curl http://localhost:3001/api/schools
```

```

[
  {"_id":"665f1fa4a7d3f1a0aabc1001","name":"Greenwood High School","address":"123 Maple Street, Springfield","principal":"Mr. John Adams"},
  {"_id":"665f1fa4a7d3f1a0aabc1002","name":"Riverside Public School","address":"456 Oak Avenue, Riverdale","principal":"Ms. Linda Carter"},
  {"_id":"665f1fa4a7d3f1a0aabc1003","name":"Lakeside Academy","address":"789 Birch Lane, Lakeview","principal":"Dr. Maria Gomez"}
]

```

```
$ curl http://localhost:3001/api/courses
```

```

[
  {"_id":"665f1fa4a7d3f1a0aabc2001","name":"Mathematics","description":"Algebra, Geometry and Trigonometry","schoolId":{"_id":"665f1fa4a7d3f1a0aabc1001","name":"Greenwood High School", ...}},
  {"_id":"665f1fa4a7d3f1a0aabc2002","name":"Biology","description":"Study of living organisms","schoolId":{"...}},
  {"_id":"665f1fa4a7d3f1a0aabc2003","name":"History","description":"World and Indian history overview","schoolId":{"...}}
]

```

```
$ curl http://localhost:3001/api/enrollments
```

```

[
  {"_id":"665f1fa4a7d3f1a0aabc3001","studentName":"Alice Johnson","courseId":{"...","name":"Mathematics"},"enrollmentDate":"2025-06-01T00:00:00.000Z"},
  {"_id":"665f1fa4a7d3f1a0aabc3002","studentName":"Bob Williams","courseId":{"...","name":"History"},"enrollmentDate":"2025-06-04T00:00:00.000Z"}
]

```

Note: the provided server.js runs on port 3001 (not 5000) and mounts routes at /api/schools, /api/courses, /api/enrollments (there is no /api/users route) - this is the actual behaviour of the provided starter code, which differs slightly from the port/route example mentioned in the narrative instructions text.